Why Minimum-Snap x_d and u_d Can Grow for Nearby Waypoints

Actual-data experiment

This report uses actual verified RRT and LLM waypoints from calibration sample 105; it does not synthesize a new prediction. The six LLM segment lengths range from 0.505 m to 0.749 m. The experiment keeps those waypoint locations fixed and uniformly multiplies all segment durations by $\rho \in \{1, 0.75, 0.5, 0.25\}$. Thus any increase in state derivatives or control is caused by time parameterization, not greater waypoint distance.

ρ	horizon [s]	s_x	s_u	max $\|v\|$	max $\|a\|$	max $\|p^{(4)}\|$	max $\|u\|$
1.00	8.610	0.656	1.228	0.810	1.255	11.968	1.268
0.75	6.457	0.837	2.121	1.079	2.230	37.824	2.000
0.50	4.305	1.247	4.634	1.619	5.018	191.484	3.973
0.25	2.152	2.486	17.981	3.238	20.072	3063.749	16.742

Time normalization and coefficient growth

Write one seventh-degree segment using normalized time $\sigma = t/T$:

$$p(t) = \sum_{k=0}^{7} a_k \sigma^k = \sum_{k=0}^{7} \underbrace{\frac{a_k}{T^k}}_{c_k} t^k.$$

The physical-time coefficient is therefore $c_k = a_k/T^k$. In particular, the fourth through seventh coefficients contain $T^{-4}, T^{-5}, T^{-6}, T^{-7}$. This is coefficient growth under a change of coordinates; it can also worsen numerical conditioning when T is small.

Every derivative adds another inverse power of duration:

$$\frac{d^r p}{dt^r} = T^{-r} \frac{d^r p}{d\sigma^r}.$$

Velocity scales as T^{-1}, acceleration as T^{-2}, jerk as T^{-3}, and snap as T^{-4}. The minimum-snap objective itself scales as

$$\int_0^T \|p^{(4)}(t)\|^2 dt = T^{-7} \int_0^1 \left\| \frac{d^4 p}{d\sigma^4} \right\|^2 d\sigma.$$

Minimum snap minimizes this large quantity subject to waypoint and continuity constraints; it cannot make the required derivatives small when the assigned time is too short.

How oscillation is amplified

A small between-waypoint ripple $e(t) = A\sin(\omega t)$ has derivatives $A\omega^r$. Its position amplitude remains A, but its acceleration is $A\omega^2$ and snap is $A\omega^4$. Shorter segment time raises the effective frequency. High-order coupled fits can also overshoot between constraints even when adjacent waypoints are close; differentiation makes such visually small lobes significant in v, a, and snap. The reviewed linear/parabolic-blend paper explicitly motivates its lower-order construction by the computational burden and Runge-type behavior of higher-order interpolation.

For this repository,

$$x_d = [p_x, p_y, v_x, v_y]^\top, \qquad u_d = v_d + \frac{a_d}{1.1}.$$

Consequently, position does not acquire a direct T^{-4} factor. The velocity part of x_d scales as T^{-1}, while u_d contains T^{-1} and T^{-2} terms and is normally acceleration-dominated for short durations. The T^{-4} factor belongs to snap and to fourth-order differentiation, not directly to this planar control definition. For a full differentially flat quadrotor, higher derivatives can enter attitude, angular-rate, and feedforward mappings, so amplification can propagate further; this relationship is discussed in the reviewed DFBC/NMPC paper.

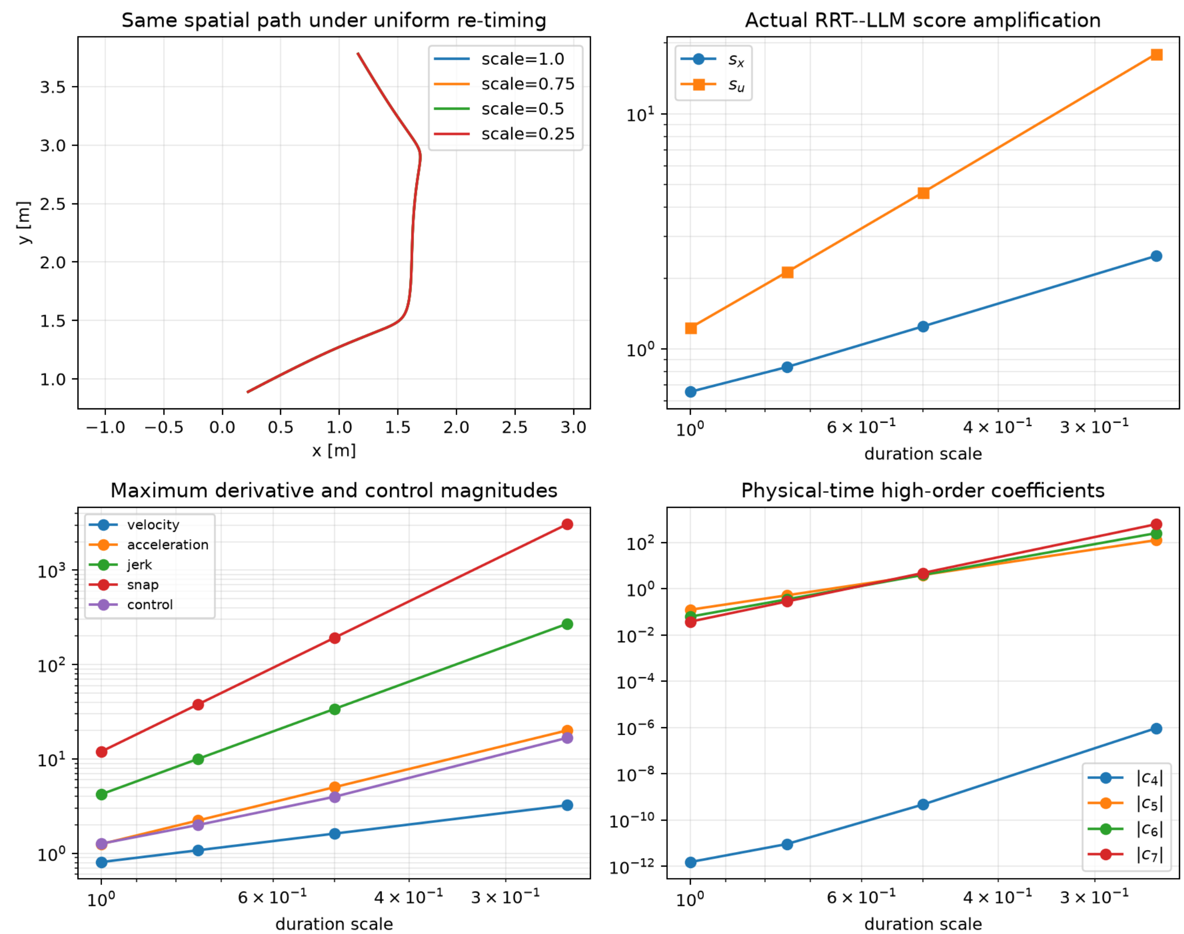

Figure 1: Actual sample 105: identical waypoint geometry, score growth, derivative growth, and physical-time coefficient growth under re-timing.

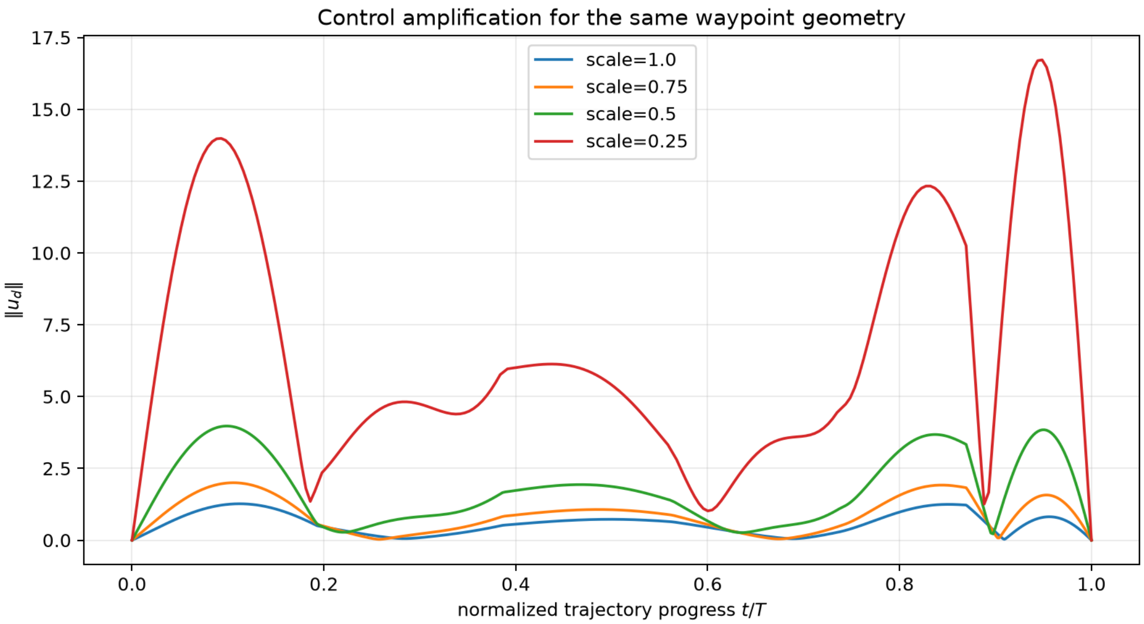

Figure 2: The same LLM waypoint path produces increasingly large control peaks as its duration is shortened.

Implication for conformal radius

The calibration scores take maxima over time. A brief derivative peak therefore becomes the sample's s_u or s_x and can enter a high conformal quantile. The contraction radius weights q_x and q_u, so a small number of short-time or oscillatory polynomial fits can enlarge the radius for every subsequent trajectory. Shared scheduling removes one source of time mismatch, Frenet matching removes wall-clock phase during comparison, and guided parabolic blends avoid high-order polynomial differentiation; none substitutes for enforcing feasible segment times and explicit derivative/input limits.